SIMATS ENGINEERING

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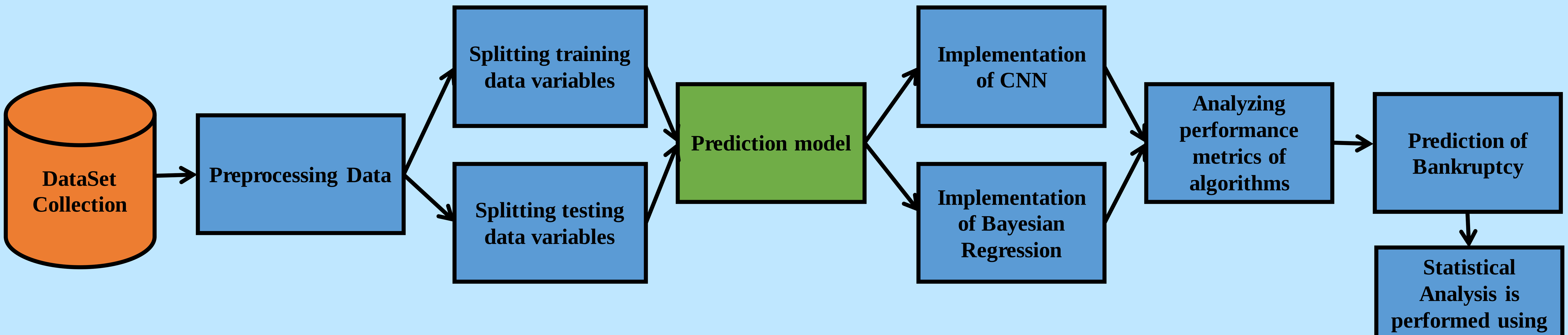
A Novel Approach for Bankruptcy Detection using Convolutional Neural Network Compared with Bayesian Regression

INTRODUCTION

- Bankruptcy is a financial crisis where a company/firm couldn't return or pay back the debt , in others words during Bankruptcy an entire company will be at Stake with no hope of surviving in the market due to the amount of loans the company has borrowed .
- The aim of this study is to identify the Bankruptcy to predict the accuracy using machine learning algorithms, thereby facilitating improvised performance of a firm and changing the financial policies to protect from future financial losses.
- This research holds importance not only for identifying Bankruptcy but also it navigates the ideology to maintain and protect a company/firm from uncertain monetary problems .
- The earlier intervention of Bankruptcy detection using machine learning algorithms helps in identifying the Bankruptcy threat and helps the company to grow sustainably while knowing it's financial status.
- The Bankruptcy data set for the study is collected form Kaggle datasets that has with several training sets of financial and monetary data required to calculate the financial stress parameters.

Bankruptcy detection via Machine Learning Algorithms

MATERIALS AND METHODS

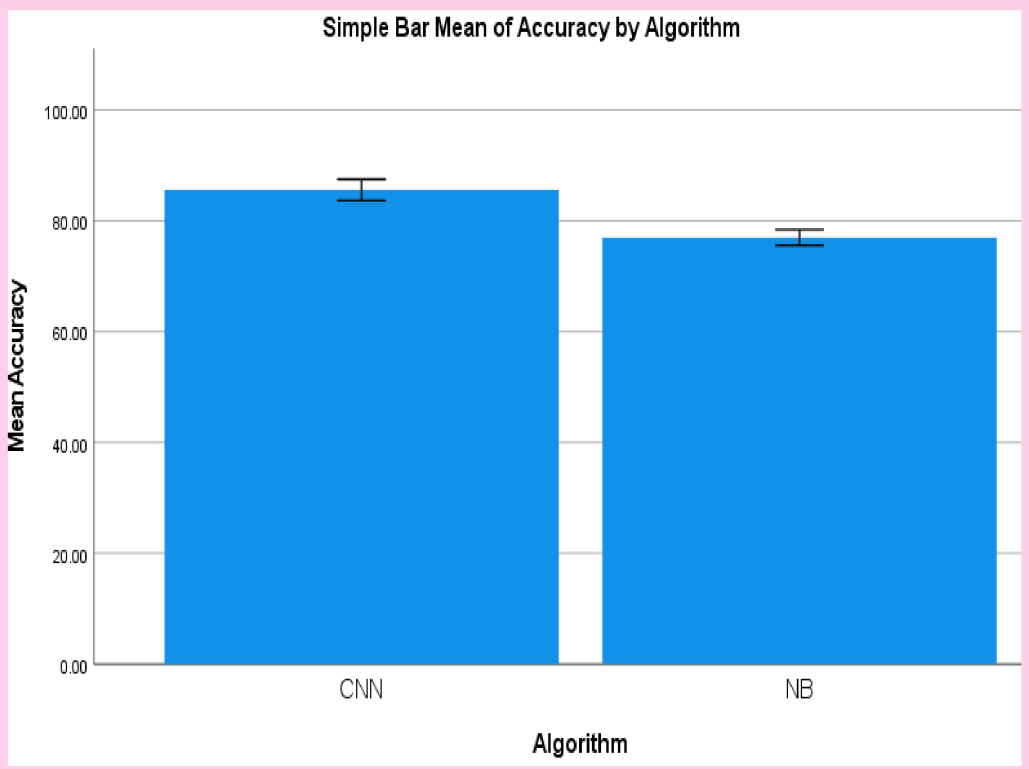


Flow Chart of Machine Learning Algorithms involved in Prediction Model

RESULTS

	GROUP	N	MEAN	Std. Deviation	Std. Error Mean
ACCURACY	CNN	10	80.4000	1.67332	0.74833
	BR	10	78.6000	2.31368	1.41339

Statistical Calculations of CNN and BR



CNN vs Bayesian Regression

Algorithm	Accuracy
CNN	87.20%
BR	74.10%

Average Accuracy values

DISCUSSION AND CONCLUSION

- Based on t-test Statistical analysis, the significance value of p=0.001 (independent sample t - test p<0.05) is obtained and shows that there is a statistical significant difference between the CNN and Bayesian Regression.
- The CNN algorithm has Mean accuracy of 80.4000% with a low standard deviation of 1.67332 and standard mean error of 0.74833 where as Bayesian Regression algorithm achieved Mean accuracy of 78.6000% with a standard deviation of 2.31368 and standard mean error of 1.41339.
- The research findings show a significant difference in migraine prediction accuracy between the Convolutional Neural Network (CNN) and Bayesian Regression (BR) algorithms, Where CNN algorithm has higher accuracy rates in terms of prediction model.
- The model's capabilities can be enhanced through the incorporation of Machine learning algorithms, facilitating real-time monitoring and predictions with greater accuracy and efficiency.
- Furthermore , the impact of this model changes the strategy of a company and their future criteria by providing them insights about the future crisis.

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